Problem Sheet: Week 10

1. Consider the "beta/binomial" model for a portfolio consisting of n independent policies, on each of which at most one claim can occur in a year. Let $\mathbf{P}$(claim) = p for each policy, where p is fixed but unknown. It follows that the total number of claims $X|p \sim$ Bin(n, p). Suppose p has a Beta(α, β) prior distribution.

   (a) Derive the posterior distribution of p and show that the posterior mean can be expressed in the form of a credibility estimate.

   (b) In the case that claims arise on 56 policies in a portfolio consisting of 10,000 policies, and we adopt a prior with mean 0.005 and standard deviation 0.0015, show that the credibility factor is $Z = 0.819$ and calculate the credibility estimate of p.

2. Suppose $X_1, X_2, \ldots, X_n$ are identically distributed random variables whose distributions depend on the value of a fixed but unknown parameter θ. The value of θ is the same for all of the X_i. Regarding θ as a random variable we have:

$$X_i|\theta \sim \mathrm{N}(\theta, \sigma_1^2), i = 1, 2, \ldots n \quad \text{and} \quad \theta \sim \mathrm{N}(\mu, \sigma_2^2)$$

   Given θ, the random variables $X_1, X_2, \ldots, X_n$ are conditionally independent.

   (a) Derive the posterior distribution of θ and show that the posterior mean can be expressed in the form of a credibility estimate.

   (b) Interpret your results. How does the credibility factor change with respect to n, σ_1^2, and σ_2^2?

3. The number of claims each year from a risk has a Poisson distribution with mean θ. The value of θ is fixed but unknown. Suppose we assume a flat prior on θ, i.e. $\pi(\theta) \propto 1$, and we take Im$(\theta) = (0, \infty)$. Let $x_1, x_2, \ldots, x_n$ be the observed numbers of claims in successive years from this risk and let

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i.$$

   (a) Show that the mean of the posterior distribution can be written as

$$\mathbf{E}\left(\theta|\underline{\mathrm{x}}\right) = \frac{\int_0^\infty \exp\{-n\theta\}\theta^{n\bar{x}+1}\, d\theta}{\int_0^\infty \exp\{-n\theta\}\theta^{n\bar{x}}\, d\theta}$$

   By using the normalising constant of the gamma distribution, show that the mean of the posterior distribution can be simplified as

$$\mathbf{E}\left(\theta|\underline{\mathrm{x}}\right) = \frac{1}{nB(1, n\bar{x}+1)}$$

   where $B(\alpha, \beta) = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha+\beta)$ is the beta function.

(b) Now suppose $n = 5$. Calculate the mean of the posterior distribution of θ for:

i. $\bar{x} = 5$,

ii. $\bar{x} = 6.4$, and

iii. $\bar{x} = 8.7$ [**Hint:** You may wish to use numerical computations in R].

(c) Show that the posterior mean can be further simplified to

$$\mathbf{E}(\theta|\underline{\mathbf{x}}) = \bar{x} + \frac{1}{n}$$

$\left[\textbf{Hint:}\ \text{The beta function is defined as } B(\alpha,\beta) = \int_0^1 t^{\alpha-1}(1-t)^{\beta-1}dt\right]$

4. A non-life insurance company has six separate portfolios, each of which has been in existence for eight years. Data on the mean and variance of the claim amounts (adjusted for inflation) are given in the table:

Portfolio number	Within-portfolio mean	Within-portfolio variance
1	87	692
2	100	1361
3	135	1015
4	112	638
5	116	859
6	110	841

Denote by θ the risk parameter of an arbitrary portfolio. Solve the following tasks:

(a) Estimate $\mathbf{E}[m(\theta)]$, the mean of the distribution of the portfolio means.

(b) Estimate $\mathbf{E}[s^2(\theta)]$, the expected variance of the within-portfolio claim amounts.

(c) Estimate $\mathbf{Var}[m(\theta)]$, the variance of the distribution of the portfolio means.

(d) Find the credibility estimate of the pure premium for Portfolio 3 according to the EBCT Model 1.